Individual Contribution Report

Student Name: Mark Ellery Contreras

Group: SureballNa (MiniOSNova)

Assigned Role: Project Leader & Process Management Developer

What did I contribute to the project?

- Coordinated the group members, assigned module responsibilities, and tracked project progress to meet our submission requirements.
- Set up and managed the official GitHub repository (MiniOSNova) and connected it to Visual Studio Code.
- Combined all individual Python modules into the main program and verified that the final version ran smoothly without errors.
- Assisted in drafting the User Manual and Technical Documentation, and tested the system commands in the terminal.

What features did I personally develop?

- Designed the custom startup banner and the main "while True" loop for the command prompt.
- Implemented the input parser to separate commands from arguments and route them to the appropriate modules.
- Programmed the core general commands: help, clear, and time.
- Developed the process management module (processes.py) to simulate application execution using Process IDs (run, process, and kill commands).

What problems did I encounter?

- GitHub Authentication Error (403 Permission Denied) because a classmate's account was cached in Windows Credential Manager.
- Git Merge Conflict when our local and remote branches got out of sync.
- The program crashed with an IndexError when commands like run or mkdir were typed without a name.

How did I solve those problems?

- I removed the old GitHub credentials from Windows Credential Manager and logged in with my own account.
- I aborted the stuck Git merge and force-pushed our clean local code to GitHub.
- I added an "if len(user_input) > 1:" check to make sure the program warns the user instead of crashing when arguments are missing.

What did I learn from this project?

I learned how an operating system's command shell continuously listens for user input, parses commands, and delegates tasks to specialized modules. I also gained hands-on experience managing a team repository with Git and GitHub in VS Code, resolving merge conflicts, and organizing technical documentation.

Evidence of my contribution:

- GitHub Repository Commits: Commit history under markelleryc-cpu in https://github.com/markelleryc-cpu/MiniOSNova.
- Source Code: Complete implementation of main.py and processes.py in the project repository.
- Project Documentation: Exported PDFs in the docs/ folder and terminal test screenshots.

